
RoFormer: ENHANCED TRANSFORMER WITH ROTARY POSITION EMBEDDING

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ABSTRACT

Position encoding recently has shown effective in the transformer architecture. It enables valuable supervision for dependency modeling between elements at different positions of the sequence. In this paper, we first investigate various methods to integrate positional information into the learning process of transformer-based language models. Then, we propose a novel method named Rotary Position Embedding(RoPE) to effectively leverage the positional information. Specifically, the proposed RoPE encodes the absolute position with a rotation matrix and meanwhile incorporates the explicit relative position dependency in self-attention formulation. Notably, RoPE enables valuable properties, including the flexibility of sequence length, decaying inter-token dependency with increasing relative distances, and the capability of equipping the linear self-attention with relative position encoding. Finally, we evaluate the enhanced transformer with rotary position embedding, also called RoFormer, on various long text classification benchmark datasets. Our experiments show that it consistently overcomes its alternatives. Furthermore, we provide a theoretical analysis to explain some experimental results. RoFormer is already integrated into Huggingface: https://huggingface.co/docs/transformers/model_doc/roformer.

Keywords Pre-trained Language Models · Position Information Encoding · Pre-training · Natural Language Processing.

1 Introduction

The sequential order of words is of great value to natural language understanding. Recurrent neural networks (RNNs) based models encode tokens' order by recursively computing a hidden state along the time dimension. Convolution neural networks (CNNs) based models (CNNs) Gehring et al. [2017] were typically considered position-agnostic, but recent work Islam et al. [2020] has shown that the commonly used padding operation can implicitly learn position information. Recently, the pre-trained language models (PLMs), which were built upon the transformer Vaswani et al. [2017], have achieved the state-of-the-art performance of various natural language processing (NLP) tasks, including context representation learning Devlin et al. [2019], machine translation Vaswani et al. [2017], and language modeling Radford et al. [2019], to name a few. Unlike, RNNs and CNNs-based models, PLMs utilize the self-attention mechanism to semantically capture the contextual representation of a given corpus. As a consequence, PLMs achieve a significant improvement in terms of parallelization over RNNs and improve the modeling ability of longer intra-token relations compared to CNNs¹.

¹A stack of multiple CNN layers can also capture longer intra-token relation, here we only consider single layer setting.

The authors of He et al. [2020] argued that the relative positions of two tokens could only be fully modeled using the middle two terms of Equation (6). As a consequence, the absolute position embeddings $\mathbf{p}_m$ and $\mathbf{p}_n$ were simply replaced with the relative position embeddings $\tilde{\mathbf{p}}_{m-n}$:

$$\mathbf{q}_m^\top \mathbf{k}_n = \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n + \mathbf{x}_m^\top \mathbf{W}_q^\top \mathbf{W}_k \tilde{\mathbf{p}}_{m-n} + \tilde{\mathbf{p}}_{m-n}^\top \mathbf{W}_q^\top \mathbf{W}_k \mathbf{x}_n \quad (10)$$

A comparison of the four variants of the relative position embeddings Radford and Narasimhan [2018] has shown that the variant similar to Equation (10) is the most efficient among the other three. Generally speaking, all these approaches attempt to modify Equation (6) based on the decomposition of Equation (3) under the self-attention settings in Equation (2), which was originally proposed in Vaswani et al. [2017]. They commonly introduced to directly add the position information to the context representations. Unlikely, our approach aims to derive the relative position encoding from Equation (1) under some constraints. Next, we show that the derived approach is more interpretable by incorporating relative position information with the rotation of context representations.

3 Proposed approach

In this section, we discuss the proposed rotary position embedding (RoPE). We first formulate the relative position encoding problem in Section (3.1), we then derive the RoPE in Section (3.2) and investigate its properties in Section (3.3).

3.1 Formulation

Transformer-based language modeling usually leverages the position information of individual tokens through a self-attention mechanism. As can be observed in Equation (2), $\mathbf{q}_m^\top \mathbf{k}_n$ typically enables knowledge conveyance between tokens at different positions. In order to incorporate relative position information, we require the inner product of query $\mathbf{q}_m$ and key $\mathbf{k}_n$ to be formulated by a function g , which takes only the word embeddings $\mathbf{x}_m, \mathbf{x}_n$, and their relative position $m - n$ as input variables. In other words, we hope that the inner product encodes position information only in the relative form:

$$\langle f_q(\mathbf{x}_m, m), f_k(\mathbf{x}_n, n) \rangle = g(\mathbf{x}_m, \mathbf{x}_n, m - n). \quad (11)$$

The ultimate goal is to find an equivalent encoding mechanism to solve the functions $f_q(\mathbf{x}_m, m)$ and $f_k(\mathbf{x}_n, n)$ to conform the aforementioned relation.

3.2 Rotary position embedding

3.2.1 A 2D case

We begin with a simple case with a dimension $d = 2$. Under these settings, we make use of the geometric property of vectors on a 2D plane and its complex form to prove (refer Section (3.4.1) for more details) that a solution to our formulation Equation (11) is:

$$\begin{aligned} f_q(\mathbf{x}_m, m) &= (\mathbf{W}_q \mathbf{x}_m) e^{im\theta} \\ f_k(\mathbf{x}_n, n) &= (\mathbf{W}_k \mathbf{x}_n) e^{in\theta} \\ g(\mathbf{x}_m, \mathbf{x}_n, m - n) &= \text{Re}[(\mathbf{W}_q \mathbf{x}_m)(\mathbf{W}_k \mathbf{x}_n)^* e^{i(m-n)\theta}] \end{aligned} \quad (12)$$

where $\text{Re}[\cdot]$ is the real part of a complex number and $(\mathbf{W}_k \mathbf{x}_n)^*$ represents the conjugate complex number of $(\mathbf{W}_k \mathbf{x}_n)$. $\theta \in \mathbb{R}$ is a preset non-zero constant. We can further write $f_{\{q,k\}}$ in a multiplication matrix:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \begin{pmatrix} \cos m\theta & -\sin m\theta \\ \sin m\theta & \cos m\theta \end{pmatrix} \begin{pmatrix} W_{\{q,k\}}^{(11)} & W_{\{q,k\}}^{(12)} \\ W_{\{q,k\}}^{(21)} & W_{\{q,k\}}^{(22)} \end{pmatrix} \begin{pmatrix} x_m^{(1)} \\ x_m^{(2)} \end{pmatrix} \quad (13)$$

where $(x_m^{(1)}, x_m^{(2)})$ is $\mathbf{x}_m$ expressed in the 2D coordinates. Similarly, g can be viewed as a matrix and thus enables the solution of formulation in Section (3.1) under the 2D case. Specifically, incorporating the relative position embedding is straightforward: simply rotate the affine-transformed word embedding vector by amount of angle multiples of its position index and thus interprets the intuition behind *Rotary Position Embedding*.

3.2.2 General form

In order to generalize our results in 2D to any $\mathbf{x}_i \in \mathbb{R}^d$ where d is even, we divide the d -dimension space into $d/2$ sub-spaces and combine them in the merit of the linearity of the inner product, turning $f_{\{q,k\}}$ into:

$$f_{\{q,k\}}(\mathbf{x}_m, m) = \mathbf{R}_{\Theta, m}^d \mathbf{W}_{\{q,k\}} \mathbf{x}_m \quad (14)$$

where

$$\mathbf{R}_{\Theta, m}^d = \begin{pmatrix} \cos m\theta_1 & -\sin m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ \sin m\theta_1 & \cos m\theta_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cos m\theta_2 & -\sin m\theta_2 & \cdots & 0 & 0 \\ 0 & 0 & \sin m\theta_2 & \cos m\theta_2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & \cos m\theta_{d/2} & -\sin m\theta_{d/2} \\ 0 & 0 & 0 & 0 & \cdots & \sin m\theta_{d/2} & \cos m\theta_{d/2} \end{pmatrix} \quad (15)$$

is the rotary matrix with pre-defined parameters $\Theta = \{\theta_i = 10000^{-2(i-1)/d}, i \in [1, 2, \dots, d/2]\}$. A graphic illustration of RoPE is shown in Figure (1). Applying our RoPE to self-attention in Equation (2), we obtain:

$$\mathbf{q}_m^\top \mathbf{k}_n = (\mathbf{R}_{\Theta, m}^d \mathbf{W}_q \mathbf{x}_m)^\top (\mathbf{R}_{\Theta, n}^d \mathbf{W}_k \mathbf{x}_n) = \mathbf{x}^\top \mathbf{W}_q \mathbf{R}_{\Theta, n-m}^d \mathbf{W}_k \mathbf{x}_n \quad (16)$$

where $\mathbf{R}_{\Theta, n-m}^d = (\mathbf{R}_{\Theta, m}^d)^\top \mathbf{R}_{\Theta, n}^d$. Note that $\mathbf{R}_{\Theta}^d$ is an orthogonal matrix, which ensures stability during the process of encoding position information. In addition, due to the sparsity of $\mathbf{R}_{\Theta}^d$, applying matrix multiplication directly as in Equation (16) is not computationally efficient; we provide another realization in theoretical explanation.

In contrast to the additive nature of position embedding method adopted in the previous works, i.e., Equations (3) to (10), our approach is multiplicative. Moreover, RoPE naturally incorporates relative position information through rotation matrix product instead of altering terms in the expanded formulation of additive position encoding when applied with self-attention.

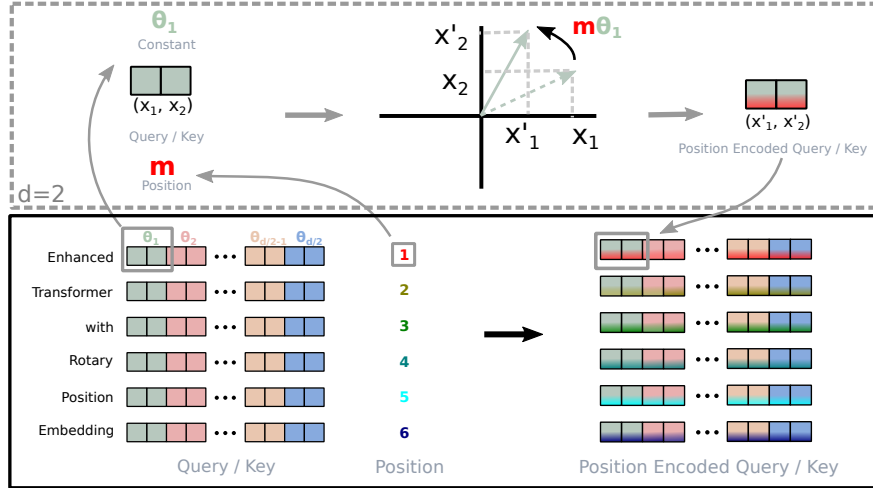


Figure 1: Implementation of Rotary Position Embedding(RoPE).

3.3 Properties of RoPE

Long-term decay: Following Vaswani et al. [2017], we set $\theta_i = 10000^{-2i/d}$. One can prove that this setting provides a long-term decay property (refer to Section (3.4.3) for more details), which means the inner-product will decay when the relative position increase. This property coincides with the intuition that a pair of tokens with a long relative distance should have less connection.

RoPE with linear attention: The self-attention can be rewritten in a more general form.